Neural Networks-2

★ Softmax Classifier (Activation Function):-

An activation func. used in NN to compare all the classes at the same (giving high prob to one class the reducing the prob. of other classes) ensuring that the class prob. sum upto 1.

It is written as -

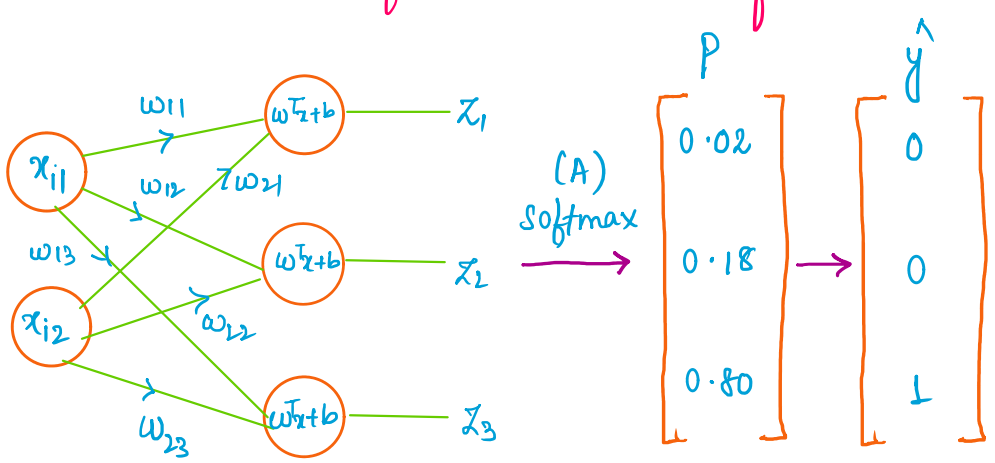
$$P_i = \frac{e^{z_i}}{\sum_{i=0}^{K} e^{z_i}}$$

where, P_i = prob. of datapoint belonging to class i

$\sum_{i=0}^{K} e^{z_i}$ = Normalization term

$(P_1 + P_2 + P_3 + \dots) = 1$

Softmax is best suited for multiclass classification.



★ Vectorized Implementation of NN:-

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$$\underbrace{\begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \\ \vdots & \vdots \end{bmatrix}}_{m \times 2(d)} \cdot \underbrace{\begin{bmatrix} w_{11} & w_{12} & w_{13} \\ w_{21} & w_{22} & w_{23} \end{bmatrix}}_{2 \times 3} \Longrightarrow \underbrace{\begin{bmatrix} - & - & - \\ - & - & - \\ \vdots & \vdots & \vdots \end{bmatrix}}_{m \times n}$$

$m \times 2(d)$ → features

2×3: (d) (n) → no. of neurons

$$\underbrace{\begin{bmatrix} - & - & - \\ - & - & - \\ \vdots & \vdots & \vdots \end{bmatrix}}_{m \times n} + \underbrace{\begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}}_{1 \times 3(n)} \Longrightarrow \underbrace{\begin{bmatrix} z'_{11} & z'_{12} & z'_{13} \\ z'_{21} & z'_{22} & z'_{23} \\ \vdots & \vdots & \vdots \end{bmatrix}}_{m \times n}$$

{bias matrix is broadcasted}

∴ equation will look like this -

$$\boxed{X' \cdot W'^{T} + b' = Z'}$$

And,

$$\boxed{\text{Probability} = A' = \text{softmax}(Z')}$$

* Loss function in NN:-

Loss function for Log. Reg → Log-loss [Binary cross entropy]

For multiclass classification → Categorical Cross E...

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$$\boxed{\text{Categorical Cross Entropy} = -\sum_{j=1}^{k} y_{ij} \log(P_{ij})}$$

if $k=2$

$$CE = -\left[y_{i1} \log(P_{i1}) + y_{i2} \log(P_{i2})\right]$$

↳ This is same as binary cross entropy

Steps :-

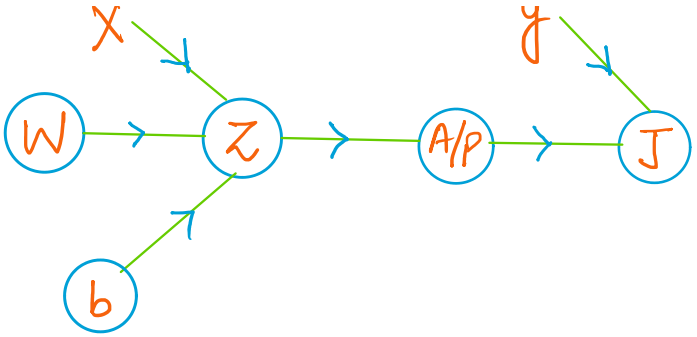
1. Initialize weight & bias matrices
2. Calculate O/P matrix
3. Calculate loss/error (J)
4. Repeat until J converge -

   update w & b :

$$\boxed{w_i = w_i - lr. \frac{\partial J}{\partial w_i}}$$

$$\boxed{b_i = b_i - lr. \frac{\partial J}{\partial b_i}}$$

★ Back Propagation :-



It is the process of computing gradients of the Loss functions w.r.t. to the weights and biases.

$$\frac{\partial J}{\partial w}, \quad \frac{\partial J}{\partial b}$$

J is not directly dependent on W & b

So, we apply chain rule here -

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial z} \cdot \frac{\partial z}{\partial w} \quad \text{—— (1)}$$

or

$$\frac{\partial J}{\partial w} = \frac{\partial J}{\partial P} \cdot \frac{\partial P}{\partial z} \cdot \frac{\partial z}{\partial w}$$

$$\left\{ \frac{10}{9} = \frac{10}{\cancel{4}} \times \frac{\cancel{4}}{9} \right\}$$

$$\frac{10}{9} = \frac{\cancel{40}^{\,10}}{\cancel{36}_{\,9}}$$

Let $\frac{\partial J}{\partial w} = dW$ & $\frac{\partial J}{\partial z} = dz$

So, eq. (1) will become

$$dW = dz \cdot \frac{\partial z}{\partial w} \quad \text{—— (2)}$$

from log. reg. $\frac{\partial z}{\partial w} = \frac{\partial [w^T x + b]}{\partial w} = x$ —— (3)

$$\& \quad \frac{\partial z}{\partial b} = \frac{\partial [w^T x + b]}{\partial b} = 1 \quad \text{— (4)}$$

And, $$dz = P_k - I(i=k) \quad \text{— (5)}$$

Here, I = indicator function

P_k = prob. of class k

Substitute the value of eq. (5) & (3) in eq. (2)

$$\boxed{dW = [P_k - I(i=k)] \cdot x} \quad \text{— (A)}$$

This is the final equation of the loss function for multiclass classification using NN.

- **Indicator function :-**

A function that returns 1 for True class labels and 0 otherwise. This is done to penalize incorrect predictions while encouraging the prob. of the correct class to be maximized.

Ex – $$\text{probs} = \begin{array}{ccc} 0 & 1 & 2 \longrightarrow \text{class labels} \\ [0.2 & 0.3 & 0.5] \end{array}$$

and True class label = 1 then according to indicator func.

$$[0.2-0 \quad 0.3-1 \quad 0.5-0]$$

$$[\underline{0.2} \quad -0.7 \quad \underline{0.5}]$$

$$\left\{ \begin{aligned} w^*_{new} &= w_{old} - [\text{gradient}] \\ &= w_{old} - (-ve) = w_{old} + _ \end{aligned} \right.$$

$\downarrow$ inc. the

inc. the wt. of correct class

Similarly, if we talk about loss w.r.t bias

let, $\frac{\partial J}{\partial b} = db$

$$db = \underbrace{\frac{\partial J}{\partial P} \cdot \frac{\partial P}{\partial z}}_{dz} \cdot \frac{\partial z}{\partial b}$$

So,

$$db = dz \cdot \frac{\partial z}{\partial b} \qquad \left[\frac{\partial z}{\partial b} = 1\right]$$

$\therefore$

$$\boxed{db = dz} \quad \text{— (B)}$$

$\therefore$ Eq. (A) & (B) are the final equations.